

Gamma delta T cells in acute myeloid leukemia: biology and emerging therapeutic strategies

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ABSTRACT

$\gamma\delta$ T cells play an important role in disease control in acute myeloid leukemia (AML) and have become an emerging area of therapeutic interest. These cells represent a minor population of T lymphocytes with intrinsic abilities to recognize antigens in a major histocompatibility complex-independent manner and functionally straddle the innate and adaptive immunity interface. AML shows high expression of phosphoantigens and UL-16 binding proteins that activate the V δ 2 and V δ 1 subtypes of $\gamma\delta$ T cells, respectively, leading to $\gamma\delta$ T cell-mediated cytotoxicity. Insights from murine models and clinical data in humans show improved overall survival, leukemia-free survival, reduced risk of relapse, enhanced graft-versus-leukemia effect, and decreased graft-versus-host disease in patients with AML who have higher reconstitution of $\gamma\delta$ T cells following allogeneic hematopoietic stem cell transplantation. Clinical trials leveraging $\gamma\delta$ T cell biology have used unmodified and modified allogeneic cells as well as bispecific engagers and monoclonal antibodies. In this review, we discuss $\gamma\delta$ T cells' biology, roles in cancer and AML, and mechanisms of immune escape and antileukemia effect; we also discuss recent clinical advances related to $\gamma\delta$ T cells in the field of AML therapeutics.

INTRODUCTION

Acute myeloid leukemia (AML) is a heterogeneous hematological malignancy that is characterized by differentiation arrest and clonal proliferation of malignant myeloid progenitor cells. Adverse-risk and relapsed/refractory (R/R) AML remain highly challenging to treat and to target in the process of drug development.^{1 2} While conventional intensive chemotherapy remains the mainstay for younger and fit patients, it may not be suitable for the majority of patients, particularly patients over the age of 60 years and those who are unfit for intensive therapy.

Venetoclax-based lower-intensity regimens have improved outcomes and are the current standard for older patients. However, cures are elusive, and the 3-year overall survival (OS) remains poor, at 25%, with dismal outcomes after the failure of frontline therapy.^{3–8}

Long-standing knowledge about the graft-vs-leukemia effect following allogeneic stem cell transplantation and transformative advances with $\alpha\beta$ chimeric antigen receptor (CAR)-T cell therapies in lymphoma and myeloma have fueled the search for effective cellular therapies for AML.^{9–11} However, these therapies have only been modestly successful, with none receiving regulatory approval so far. Toxicities noted with CAR-T cell therapies, including cytokine release syndrome (CRS) and immune effector cell-associated neurotoxicity syndrome (ICANS), have led to increased interest in alternative strategies with a lower risk of such toxicities, including natural killer (NK) cell-based therapies.

NK cell-based and $\gamma\delta$ T cell-based strategies have emerged as a promising new approach to counter AML because of these cells' unique biology, lower secretion of cytokines that are implicated in CRS and ICANS, and low risk of graft-vs-host disease (GvHD). $\gamma\delta$ T cells are a subset of T lymphocytes that express the $\gamma\delta$ T-cell receptor (TCR) instead of the $\alpha\beta$ TCR, which is expressed by the $\alpha\beta$ T cells that form the majority of T lymphocytes in humans. In this review, we discuss $\gamma\delta$ T cells' biology, roles in cancer and AML, and mechanisms of immune escape and antileukemia effect; we also discuss recent clinical advances using $\gamma\delta$ T cells in the field of AML therapeutics.

Development of $\gamma\delta$ T cells

$\gamma\delta$ T cells develop in the thymus during fetal maturation.¹² The double-negative thymocytes (CD4[−] and CD8[−]), through multiple developmental stages and gene rearrangements, result in the emergence of two primary T cell lineages, $\alpha\beta$ and $\gamma\delta$ T cells.¹³ According to the Signal-Strength Model, the intensity of the TCR signal to the double-negative thymocytes, which is dependent on the extracellular signal-regulated kinase pathway, determines the commitment of these cells to the $\alpha\beta$ or $\gamma\delta$ T cell lineages, with a strong signal, among



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Table 1 Key differences between $\alpha\beta$ T cells, $\gamma\delta$ T cells, iNKT cells, and NK cells

Properties	$\gamma\delta$ T cells	$\alpha\beta$ T cells	NK cells	iNKT cells
Surface receptor variety	$\gamma\delta$ TCR	$\alpha\beta$ TCR	AR, IR	invariant TCR
Frequency in peripheral blood (% of total lymphocytes)	0.3–3.35	63–66	5–15	0.01–0.36
Distribution	Epithelial layers, blood	Lymphoid organs, tissues	Blood, lymphoid organs, tissues	Thymus, liver
Type of immunity	Innate+adaptive	Adaptive	Innate	Innate
Require antigen presentation	No	Yes	No	No
Repertoire diversity	Limited	Almost unlimited	Limited	Limited
Effector cell generation time	1–2 days	7 days	3 days	–
Predominant cytometric phenotype	CD4 [−] , CD8 [−]	CD4 ⁺ , CD8 ⁺	CD56, CD16, NKp46	CD4 ⁺ , CD8 ⁺ , CD24 ⁺ , CD44 ⁺
GvHD risk	Minimal	Yes	No	Minimal
Risk of CRS/ICANS	Low	Yes	Low	Low

AR, activating receptor; CRS, cytokine release syndrome; GvHD, graft-versus-host disease; ICANS, immune effector cell-associated neurotoxicity syndrome; IR, inhibitory receptor; NK, natural killer; TCR, T cell receptor.

other factors, favoring the $\gamma\delta$ T cell lineage.¹⁴ The double-positive thymocytes (CD4⁺, CD8⁺) give rise to multiple cell lineages, including cells that express the V α 14+ invariant $\alpha\beta$ TCR, which act as progenitors of invariant NK T cells.¹⁵ The key differences among the $\alpha\beta$ T cell, $\gamma\delta$ T cell, invariant NK T cell, and NK cell lineages are summarized in [table 1](#).

$\gamma\delta$ T cells can be classified on the basis of TCR V δ expression as V δ 1, V δ 2, V δ 3, and V δ 5 subtypes.¹² V δ 1 and V δ 3 T cells are present in the intestinal mucosa, skin, liver, lungs, reproductive tract, thymus, and spleen, while V δ 2 T cells are predominantly found in the peripheral blood.^{16–19} On the basis of TCR V γ expression, there are numerous $\gamma\delta$ T cell types, V γ 2–9 and V γ 11.^{16–20} $\gamma\delta$ T cells can differentiate between healthy and tumor cells through their reliance on certain physiological receptors, such as the $\gamma\delta$ TCR and the natural cytotoxicity receptors, including NKp30 and NKp44.^{21–24}

Healthy adults have approximately a median of 63 $\gamma\delta$ T cells/ μ L (9–253 cells/ μ L), representing 4.3% (1.2%–15.4%) of all T cells in the peripheral blood.²⁵ V δ 2 T cells comprise approximately 1%–10% of all T cells in the peripheral blood, with absolute counts of nearly 39 cells/ μ L (6–243 cells/ μ L),²⁵ and more than 60% of $\gamma\delta$ T cells. Of all the V δ 2 T cell subtypes, V γ 9V δ 2 occupies the largest proportion, constituting 50%–95% of V δ 2 T cells.²⁶ In contrast, V δ 1 T cells comprise only 10%–30% of $\gamma\delta$ T cells in the peripheral blood, suggesting absolute counts of nearly a median of 16 cells/ μ L (2–75 cells/ μ L).²⁵ The distribution of $\gamma\delta$ T cells in the lymphoid tissue, including the gut and skin-associated lymphoid systems, is similar to that seen in the peripheral blood.²⁷

In addition to differences in the TCR representation, $\gamma\delta$ T cells are known to differ from $\alpha\beta$ T cells in their antigen recognition. V δ 1 T cells recognize tumor cells through their interaction with major histocompatibility complex

(MHC)-related molecules, such as MHC class I-related molecule A, MHC class I-related molecule B, and UL16 binding proteins (ULBPs), or via the detection of MHC-related class-Ib molecules representing the glycolipids CD1c and CD1d.²⁸ In contrast, V δ 2 T cells detect phosphoantigens (pAgs) such as isopentenyl pyrophosphate, a derivative of the mevalonate pathway, or those released by pathogens (eg, [E]–4-hydroxy-3-methyl-but-2-enyl-pyrophosphate) and are activated by them.²⁹ Although an in-depth understanding of the mechanism is yet to be developed, V γ 9V δ 2 T cells are specifically thought to be activated through their interaction with pAgs, dependent on the butyrophilin (BTN) family (BTN3A1, BTN3A2, BTN3A3, and BTN2A1) of proteins.³⁰ V δ 3 T cells, similar to V δ 1 T cells, recognize glycolipids bound to CD1d molecules expressed by antigen-presenting cells.³¹

Functions of $\gamma\delta$ T cells

$\gamma\delta$ T cells serve as a bridge between the innate and adaptive immune systems.²⁶ As $\gamma\delta$ T cells are essentially T lymphocytes consisting of TCR rearrangements, their ability to induce immunological memory and destroy target cells categorizes them as members of the adaptive immune system.²⁶ On the other hand, their ability to recognize multiple molecular patterns, resulting in swift responses independent of MHC restriction, and the coexpression of receptors similar to those expressed by other members of the innate immune system, such as activating NK receptors (NKG2D, NKp30, and NKp44) and Toll-like receptors, are characteristics that contribute to their categorization as members of the innate immune system.²⁹

$\gamma\delta$ T cells play an important role in the physiologic functioning of the immune system. These cells induce immunity against pathogenic organisms, regulate responses of the immune system, promote tissue healing, and carry out tumor surveillance in the normal human body.²⁹

Moreover, $\gamma\delta$ T cells, like adaptive T cells, differentiate into various T helper-like cells (Th1-like, Th2-like, and Th17-like cells), leading to the production of a multitude of cytokines via which they can orchestrate the function of other innate and adaptive cells³² (figure 1). The presence of transformed, infected, or stressed cells could lead to the induction of cell death in the target cells through $\gamma\delta$ T cells, mediated by the production of cytotoxic molecules, such as perforins and granzymes, or by activating pathways involved in increasing the expression of tumor necrosis factor-related apoptosis-inducing ligand receptors and death-inducing receptors (CD95).²⁹ $\gamma\delta$ T cells also interact with neutrophils, $\alpha\beta$ T cells, dendritic cells, and B cells to exert their immune response. These cells are known to draw other immune mediators to the site of infection via the production of interferon-gamma (IFN- γ) and interleukin-17 (IL-17).²⁶

$\gamma\delta$ T cells, as tumor-infiltrating lymphocytes, are a marker of significantly favorable cancer prognosis.¹⁶ Multiple studies have suggested that they have antitumor activity in numerous solid cancers, such as melanoma, prostate cancer, bladder cancer, pancreatic cancer, colon cancer, and breast cancer and hematological malignancies such as lymphomas, leukemias, and multiple myeloma.²⁹ The proposed mechanisms of antitumor activity include direct cytotoxicity via perforins, granzymes, and tumor necrosis factor-related apoptosis-inducing ligand receptors; Fas ligand; antibody-dependent cellular cytotoxicity; and alteration of the tumor microenvironment dynamics. $\gamma\delta$ T cells exert antibody-dependent cellular cytotoxicity, leading to tumor lysis, through the expression of CD16 (Fc γ receptor III), which interacts with the target cell-bound Fc region of antibodies.^{6,20} This has been capitalized clinically, in conjugation with certain monoclonal antibodies, such as trastuzumab and rituximab, where the stimulation of V γ 9V δ 2 T cells leads to the upregulation of CD16, resulting in antibody-dependent cellular cytotoxicity-mediated tumor lysis.¹²

Role of $\gamma\delta$ T cells in AML

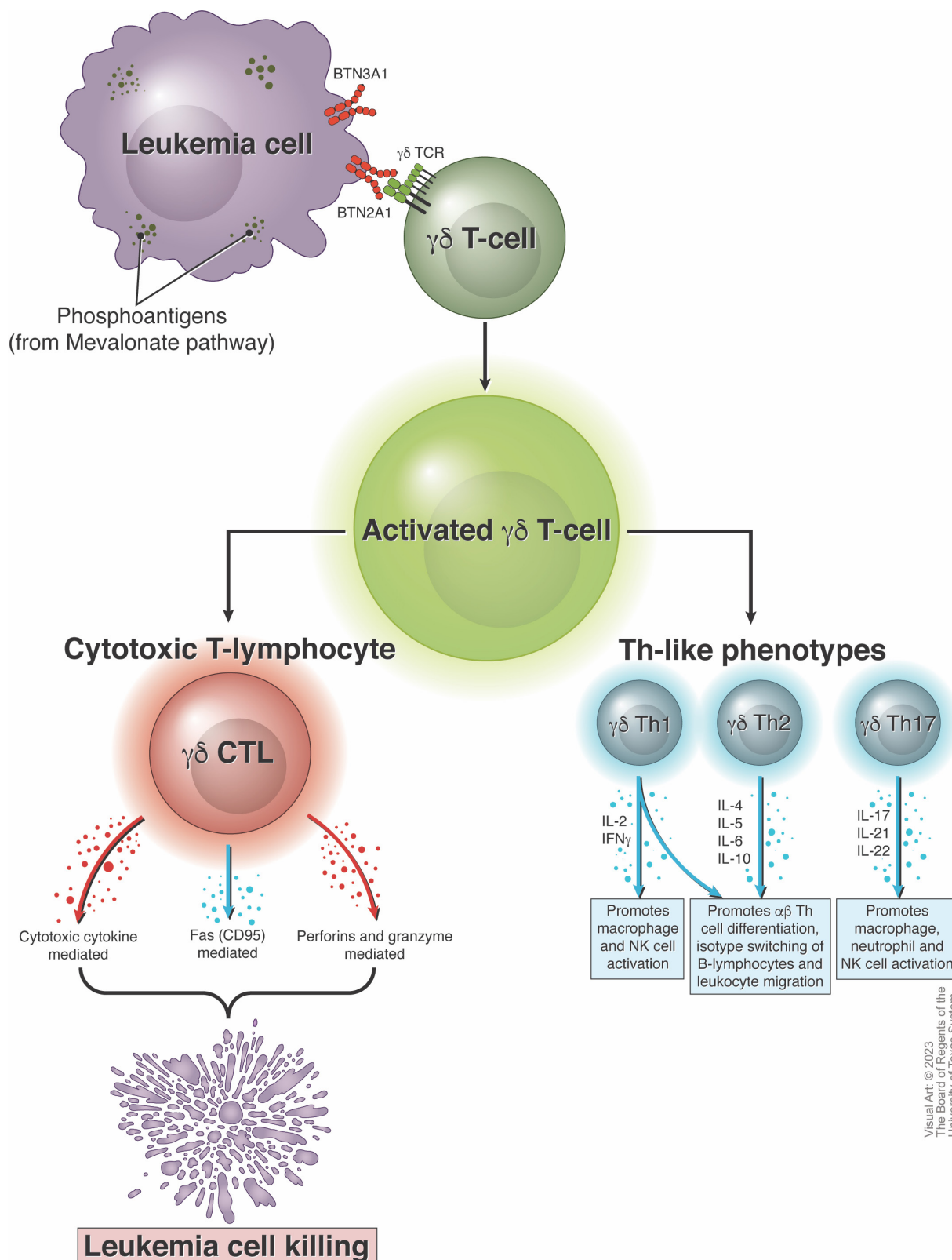
AML blasts express high levels of IL-8, CXC-chemokine ligand-10, or interferon γ -induced protein-10 and CC-motif chemokine ligand-5, also known as RANTES (regulated on activation, normal T cell-expressed and secreted), among other chemokines.^{17,18} In response to some of these factors, the tumor-reactive V δ 1 cells in the peripheral blood strongly upregulate their expression of CXC chemokine receptor-1 (CXCR1) while also increasing the expression of C-C motif chemokine receptor-2 (CCR2), CXCR3, and, minimally, CCR5, resulting in the production of higher levels of IFN- γ and homing of the V δ 1 cells into the leukemia microenvironment.^{17,18} In addition, xenotransplantation murine models have demonstrated the homing abilities of the V γ 9V δ 2 cells, from the blood into the bone marrow.¹⁸ V γ 9V δ 2 cells respond to some of these chemokines expressed in the leukemia microenvironment by strongly expressing CCR5, along with the expression of CCR3 and CXCR3

and weak expression of CXCR1.¹⁷ In addition to these mechanisms, in vitro studies have also explored the role of the CXCR4-CXC-chemokine ligand-12 pathway in the relocation of $\gamma\delta$ T cells into the bone marrow microenvironment in AML.¹⁹

$\gamma\delta$ T cells use $\gamma\delta$ TCR, coreceptors such as NK cell receptors, metabolic antigens such as pAg, and CD1-dependent mechanisms, among others, to recognize and eliminate leukemia cells.²¹ pAg are captured by BTN molecules in normal and leukemia cells. The V γ 9 chain, a component of the V γ 9V δ 2 TCR, is germline-encoded to identify BTN molecules and leads to the activation of V γ 9V δ 2 T cells.^{21,22} BTN2A1 protein has a pertinent role in the recognition of leukemia blasts, despite being incapable of directly capturing pAg molecules. It indirectly sends activating signals to the V γ 9V δ 2 T cells through the other isoforms of the BTN protein, such as BTN3A1 (CD277) and BTN3A3.²³ Moreover, the TCRs of non-V γ 9V δ 2 T cells such as V δ 1 and V δ 3 T cells, while potentially lacking pAg-BTN-V γ 9 axis recognition capabilities, are capable of more robust recognition of antigens such as pathogenic and non-pathogenic lipids when presented by a member of the CD1 protein family, particularly CD1b, CD1c, or CD1d.^{26,31} These mechanisms of antigen recognition, which are otherwise typically associated with NK cells, are a key factor in the interest in developing $\gamma\delta$ T cell-based therapies in AML and other hematological malignancies.^{21,24,31} The process of antigen recognition mediated by V δ 1 $\gamma\delta$ T cells has been shown to identify leukemia cells with upregulated expression of ULBP2, ULBP3, and MHC class I-related molecule A (figure 2), leading to higher IFN- γ and tumor necrosis factor-mediated antitumor effects, while V δ 2 $\gamma\delta$ T cells can recognize upregulated ULBP1 and ULBP2 expression.^{21,33} AML blasts express high to very high levels of ULBP1, which is a strong incentive for the development of $\gamma\delta$ T cells for AML immunotherapy.³⁴ Molecules such as nectin-2 (CD112) and polio virus receptor (CD155), expressed in leukemia cells, may also act as ligands for NK cell receptors like DNAX accessory molecule-1, which have been observed to play an advantageous role in immunotherapy for AML.^{21,35–38}

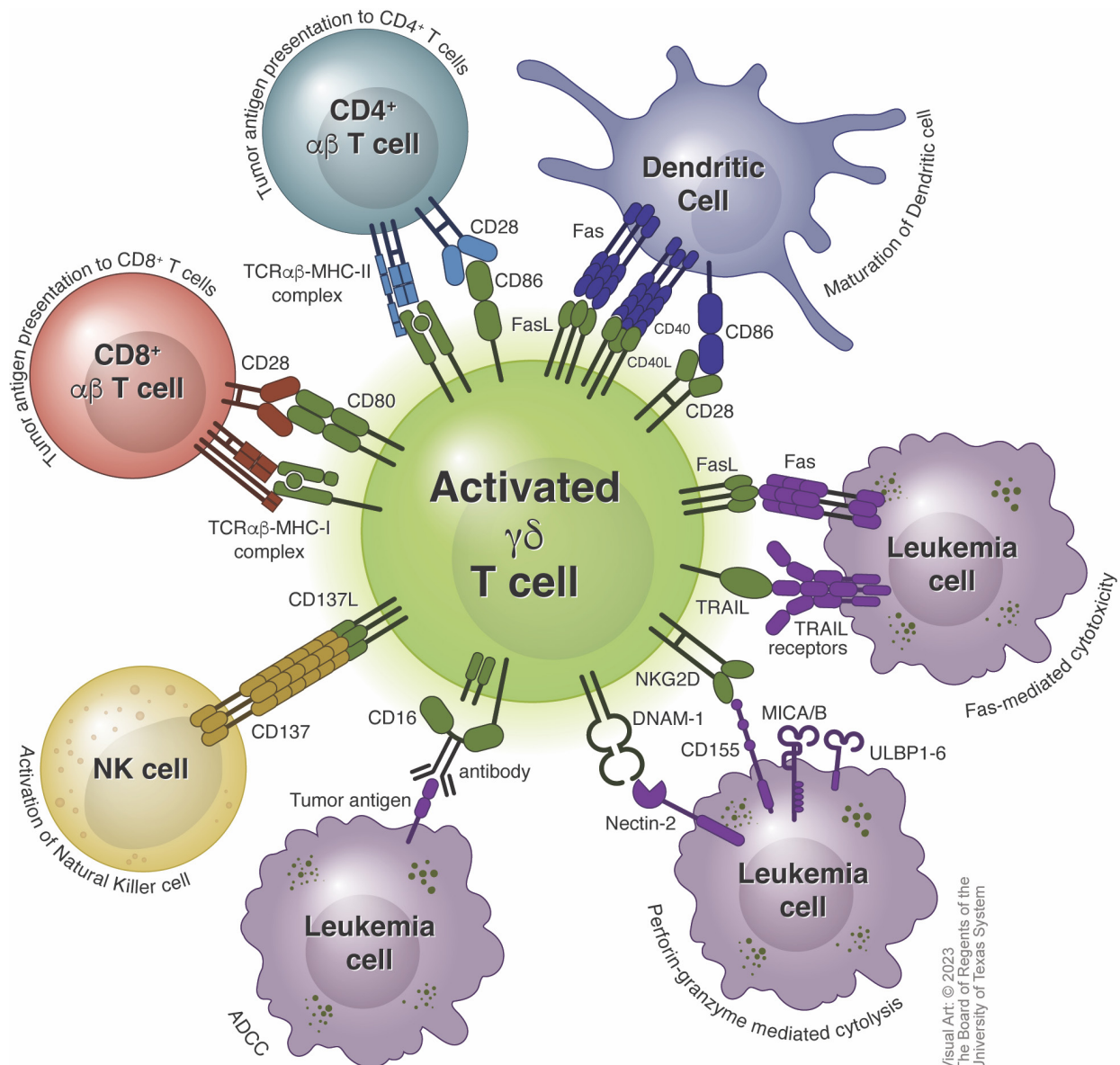
Mechanisms of immune escape from $\gamma\delta$ T cell-mediated eradication

AML blasts can evolve to select for clones that are capable of maintaining an immunosuppressive and highly tolerogenic milieu to evade $\gamma\delta$ T cell-mediated killing.³⁹ AML progenitor cells are known to upregulate regulatory T cells and myeloid-derived suppressor cells (MDSCs) and repressive factors, such as transforming growth factor-beta, adenosine, prostaglandin-E2, and enzymes such as arginase and indoleamine 2,3-dioxygenase.⁴⁰ Upregulation of immune checkpoints, such as programmed cell death protein-1 (PD-1), T cell immunoglobulin and mucin domain-containing protein 3 (TIM-3), and T cell immunoreceptor with Ig and ITIM domains (TIGIT), and epigenetic downregulation of human leucocyte



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Figure 1 Effector phenotypes of $\gamma\delta$ T cells. Butyrophilin molecules expressed on AML blasts sense phosphoantigens derived from the mevalonate pathway, activate $\gamma\delta$ T cells, and promote their differentiation into two major phenotypes—cytotoxic T lymphocytes and T helper-like lymphocytes. The cytotoxic T lymphocyte forms promote leukemic cell killing through cytotoxic cytokines, Fas (CD95), and perforins and granzyme, whereas, the T helper-like phenotypes predominantly activate and recruit other cells of the immune system by promoting the activation of macrophages, NK cells and neutrophils; differentiation of $\alpha\beta$ T cells; isotype switching of B lymphocytes; and migration of leucocytes toward the tumor microenvironment. AML, acute myeloid leukemia; BTN, butyrophilin; CD, cluster of differentiation; CTL, cytotoxic T lymphocyte; IFN, interferon; IL, interleukin; NK, natural killer; TCR, T cell receptor; Th, T helper.



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Figure 2 Activation of $\gamma\delta$ T cells and crosstalk with other immune cell populations. Activated $\gamma\delta$ T cells process tumor antigens and present those to $CD4^+$ and $CD8^+$ $\alpha\beta$ T cells. $\gamma\delta$ T cells also promote maturation of dendritic cells and activation of NK cells. Additionally, activated $\gamma\delta$ T cells mediate leukemia cell-lysis by antibody-dependent cellular cytotoxicity, perforin-granzyme mediated cytotoxicity and Fas-mediated cytotoxicity. ADCC, antibody-dependent cellular cytotoxicity; CD, cluster of differentiation; DNAM, DNAX-activating molecule; MHC, major histocompatibility complex; NK, natural killer; TCR, T cell receptor; TRAIL, Tumor necrosis factor-related apoptosis-inducing ligand; ULBP, UL16 binding protein.

antigen class II molecules, has also been implicated in immune escape^{16 38 41 42} (figure 3). Dendritic cells, macrophages, and certain immature myeloid subsets, such as MDSCs, have inhibitory action on $\gamma\delta$ T cells. M2-type macrophages and MDSCs, which have been found to be elevated in patients with AML, modify the leukemia microenvironment by producing factors such as transforming growth factor-beta and arginase.⁴⁰ AML blasts and polymorphonuclear-MDSCs may limit IFN- γ production by V γ 9V δ 2 T cells and reduce their antileukemic cytotoxicity in an arginase-I or arginase-II-dependent manner.⁴³

AML blasts express molecules, such as programmed cell death ligand-1 (PD-L1), galectin-9, CD112 and CD155,

CD80 and/or CD86, and herpes virus entry mediator, which act as ligands for immune checkpoints expressed by $\gamma\delta$ T cells, such as PD-1, TIM-3, TIGIT, cytotoxic T-lymphocyte antigen 4 and B and T lymphocyte attenuator, respectively.⁴⁴ While PD-1, TIM-3, and TIGIT are significantly expressed in $\gamma\delta$ T cells, cytotoxic T-lymphocyte antigen 4 is only rarely expressed.⁴⁵ $\gamma\delta$ T cells express PD-1, which combines with programmed cell death ligands (PDLs) to initiate an intracellular signaling cascade that restricts the activation of $\gamma\delta$ T and $\alpha\beta$ T cells. Interestingly, PDLs belong to the B7 protein family, similar to BTN-3, which activates $\gamma\delta$ T cells.⁴⁶ A transient upregulation of PD-1 has been demonstrated on highly active $\gamma\delta$ T cells on stimulation by pAgs, possibly to avoid the targeting of healthy tissues

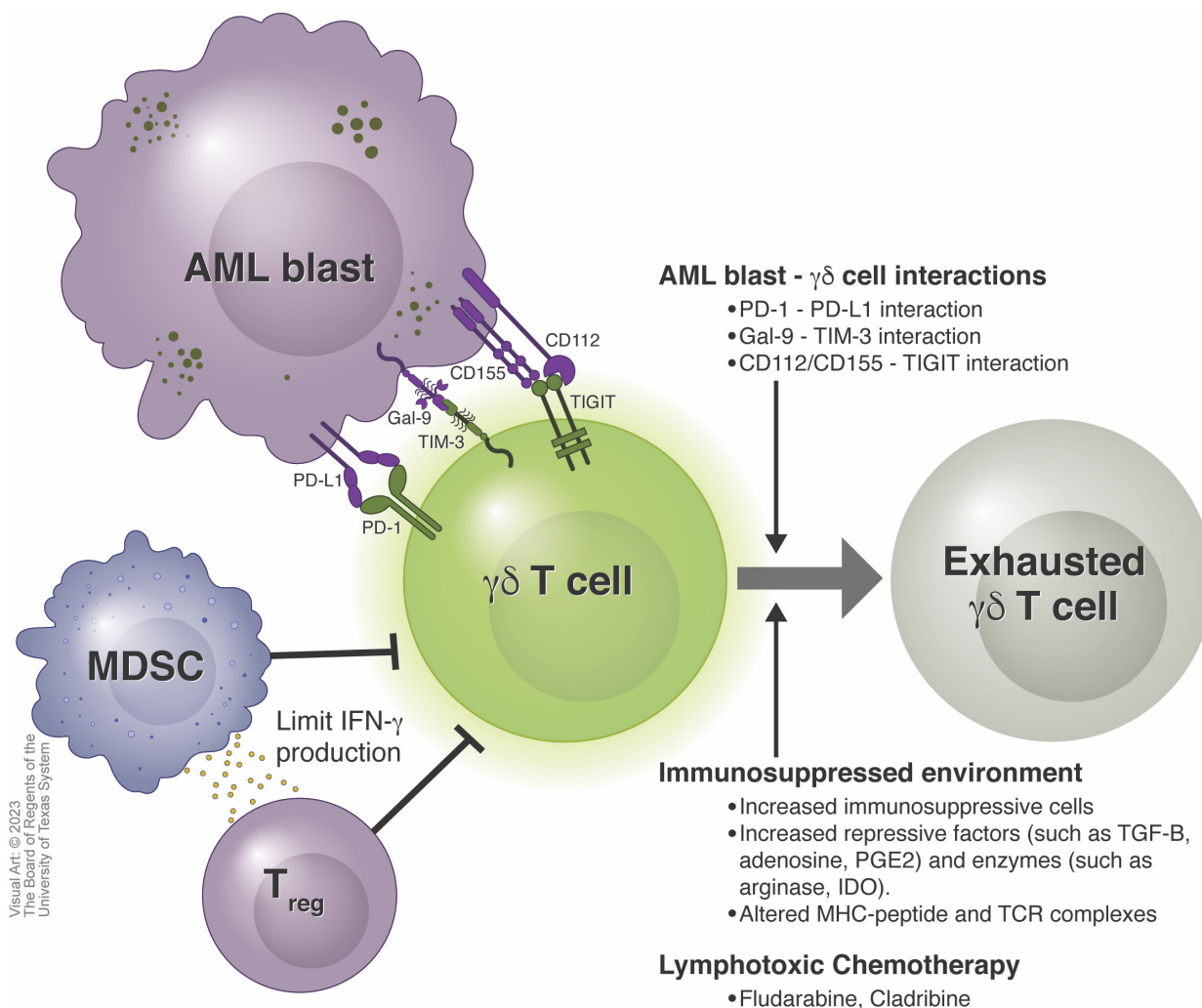


Figure 3 Factors influencing $\gamma\delta$ T cell fitness and exhaustion. Exhaustion of $\gamma\delta$ T cells can be induced in by multiple factors in the tumor microenvironment. AML blasts directly induce exhaustion of $\gamma\delta$ T cells via PD-1-PD-L1 interaction, Gal-9-TIM-3 interaction, and CD112/CD155-TIGIT interactions. An immunosuppressive milieu can accelerate exhaustion of $\gamma\delta$ T cells. Immunosuppressive MDSCs and Tregs, soluble factors, for example, TGF- β , adenosine and PGE2, and enzymes, for example, arginase and IDO, can contribute toward the immunosuppression. Lastly, lymphotoxic chemotherapy, such as fludarabine and cladribine, frequently used in AML lead to depletion and exhaustion of $\gamma\delta$ T cells. AML, acute myeloid leukemia; CD, cluster of differentiation; IDO, indoleamine 2,3-dioxygenase; MDSC, myeloid derived suppressor cells; MHC, major histocompatibility complex; PD, programmed cell death protein; PD-L1, programmed cell death protein ligand 1; PGE, prostaglandin E; TCR, T cell receptor; TGF, transforming growth factor; TIGIT, T cell immunoreceptor with Ig and ITIM domains.

that express PDL-1.⁴⁷ Simultaneous expression of PDLs, like PDL-1, has also been demonstrated on the surface of $\gamma\delta$ T cells to potentially prevent fratricide. While this may be a part of the physiological mechanism needed to prevent overactivation of the immune system, AML blasts also express the PD-1 ligands PD-L1 and PD-L2 and use this mechanism to suppress $\gamma\delta$ T cells.⁴⁰

Murine models suggest the presence of a subset of $\gamma\delta$ T cells that may demonstrate protumor activity by limiting $\alpha\beta$ T cell infiltration into leukemia sites as a consequence of PD-1/PDL-1 signaling.⁴⁶ Although the influence of PD-1/PDL-1 axis signaling on the cytotoxic potential of $\gamma\delta$ T cells is arguably weak in immunocompetent healthy environments, it may become significant in the immunosuppressed leukemia microenvironment.^{21 46} However, xenograft models indicate that $\gamma\delta$ T cells may partially

overcome this PD1/PDL-1 axis-mediated immune evasion by AML blasts if activating signals of sufficient intensity are transduced through their $\gamma\delta$ TCR.⁴⁷ Similar to PD-1, TIM-3 and TIGIT are also regulatory receptors expressed on $\gamma\delta$ T cells that can be leveraged by AML blasts for immune evasion.²¹ AML blasts express Gal-9, which acts as a ligand for TIM-3, while CD112 and CD155 are ligands for TIGIT, similar to DNAX accessory molecule-1 (CD226).^{40 48} AML patients have an increased population of bone marrow-resident V δ 1 T cells that express PD-1, TIM-3, and TIGIT.⁴⁹ The coexpression of these inhibitory molecules may suggest a state of functional exhaustion in these cells.⁵⁰ Moreover, in patients with certain subsets of AML, a higher frequency of TIGIT⁺ CD226⁻ $\gamma\delta$ T cells may be associated with an inferior 12-month OS (27% vs 81%, $p=0.028$).⁵¹ Lastly, exposure to certain types of

lymphotoxic chemotherapy, such as fludarabine and cladribine, may decrease fitness in $\gamma\delta$ T cells.

Impact of $\gamma\delta$ T cells on post-transplantation outcomes

$\gamma\delta$ T cells enhance the graft-vs-leukemia effect and reduce the risk of GvHD. Studies in murine models show that in the absence of donor $\gamma\delta$ T cells, mice had a significant reduction in body weight and survival.⁵² Death due to accelerated acute GvHD was evident in cases of $\gamma\delta$ T cell-deficient donor grafts, suggesting that donor $\gamma\delta$ T cells suppress acute GvHD, potentially through the inhibition of CD4⁺ T cell activation. Similar results have been obtained in humans in the postallogeneic transplantation setting. $\gamma\delta$ T cells reconstitute faster than $\alpha\beta$ T cells in the myeloablated marrow after transplantation. One study of patients undergoing allogeneic stem cell transplantation for myeloid and lymphoid leukemias showed significant improvement in OS and relapse-free survival, and a significantly lower risk of acute GvHD in patients with high $\gamma\delta$ T cell reconstitution at 56 days after transplantation.⁵³ A significantly lower risk of acute GvHD with death was observed in patients with higher concentrations of total $\gamma\delta$ T cells ($p=0.02$), V δ 1 $\gamma\delta$ T cells ($p=0.08$), and V δ 2 $\gamma\delta$ T cells ($p=0.02$) 28 days after allogeneic stem

cell transplantation. Patients with hematological malignancies with $\gamma\delta$ T cell concentrations lower than the median had an estimated five times higher risk of death ($HR=5.16$, $p=0.001$) and a nearly three times higher risk of relapse ($HR=2.7$, $p=0.007$) than did those whose $\gamma\delta$ T cell concentrations were higher than the median.⁵³

Among pediatric patients with leukemia undergoing allogeneic stem cell transplantation, those who recovered with a higher number of $\gamma\delta$ T cells had a significantly higher event-free survival rate than did those who recovered with a normal or lower number of $\gamma\delta$ T cells (91% vs 55%, $p=0.04$).⁵⁴ In another study in patients with acute leukemia undergoing stem cell transplantation from partially mismatched-related donors, increased $\gamma\delta$ T cells after allogeneic hematopoietic stem cell transplant (allo-HSCT) conferred significantly improved 5-year leukemia free survival (LFS) (54.4% vs 19.1%, $p<0.0003$) and OS rates (70.8% vs 19.6%, $p<0.0001$) compared with normal or lower numbers of $\gamma\delta$ T cells.⁵⁵ Importantly, recovery of $\gamma\delta$ T cells following allo-HSCT can overcome the negative prognostic impact of persistent measurable residual disease before transplantation. Newly diagnosed patients and those with R/R AML undergoing allo-HSCT

Table 2 Current clinical trials investigating the role of $\gamma\delta$ T cells in AML

Agent	Description	Ph.	Dose (cells/kg)	Population	Rec.	NCT
TCB-202-001	Ex vivo expanded allogeneic V γ 9V δ 2 T cells	I	1×10^6 1×10^7 1×10^8	R/R AML	C	NCT03790072 ⁶⁵
TCB-008	Ex vivo expanded allogeneic $\gamma\delta$ T cells	II	7×10^7 or 7×10^8	R/R AML, MRD+	R	NCT05358808 ⁶⁶
INB-100	Allogeneic $\gamma\delta$ T cells	I	1×10^6 3×10^6	R/R AML	R	NCT03533816 ⁷⁰
ICT01+VEN/AZA	V γ 9V δ 2 T cell activator (anti-BTN3A(CD 277)mAb)	II	–	Newly diagnosed AML	R	NCT04243499 ⁷⁷
$\gamma\delta$ T cell infusion	Artificial APC-expanded donor $\gamma\delta$ T cell infusion	I/Ib	1×10^6 5×10^6 2.5×10^7 1×10^8	R/R AML	R	NCT05015426 ⁷⁹
GDX012	Allogeneic $\gamma\delta$ 1 T cells	I	–	R/R AML	NYR	NCT05886491 ⁸⁰
LAVA-051*	CD1d-V δ 2TCR (V γ 9V δ 2 T cell) engager and iNKT activator (bispecific antibody)	I/IIa	100 μ g	R/R CLL, MM, AML	ANR	NCT04887259 ⁸¹
CTM-N2D	iPSC-derived $\gamma\delta$ 2 T cells (NKG2D CAR)	I	1×10^7 5×10^8 3×10^8 1×10^9	Multiple cancers	NYR	NCT05302037 ⁸²
ADI001	Allogeneic V δ 1 $\gamma\delta$ T cells (CD20 CAR)	I	–	R/R B-cell NHL, DLBCL	R	NCT04735471 ^{83 84}
ADI925	V δ 1 T cells (chimeric adapter)	–	–	Multiple cancers	–	– ⁸⁵

*Development has been discontinued.

AML, acute myeloid leukemia; ANR, active but not recruiting; APC, antigen-presenting cell(s); C, completed recruitment; CAR, chimeric antigen receptor; DLBCL, diffuse large B cell lymphoma; iPSC, induced pluripotent stem cells; mAb, monoclonal antibody; MRD, measurable residual disease; NCT, National Clinical Trial identifier; NHL, non-Hodgkin's lymphoma; NYR, not yet recruiting; Ph, phase; R, recruiting; Rec, recruitment status; R/R, relapsed/refractory; VEN/AZA, venetoclax/azacitidine.

measurable residual disease-positive remission had higher than the median level of $\gamma\delta$ T cells at 30 and 100 days after allo-HSCT and showed a significant improvement in 5-year LFS (HR=0.42, $p=0.007$ and HR=0.42, $p=0.011$, respectively) and OS (HR=0.44, $p=0.038$ and HR=0.33, $p=0.009$, respectively).⁵⁶ Additionally, patients with a lower V δ 2 T cell concentration at 90 days after haploidentical hematopoietic cell transplantation had significantly higher non-relapse mortality rates at 2 and 5 years ($p=0.017$).⁵⁷ These data suggest that overall, $\gamma\delta$ T cells exert their graft-vs-leukemia effect while mitigating acute GvHD post-allo-HSCT, leading to an improved prognosis in acute leukemia patients.

$\gamma\delta$ T cell-based adoptive cellular therapies

Robust $\gamma\delta$ T cell expansion methods are critical to advance adoptive cellular therapies using these cells. V γ 9V δ 2 T cells can be extracted from peripheral blood and expanded ex vivo owing to the distinct characteristics of the $\gamma\delta$ TCR. This has been capitalized to develop allogeneic $\gamma\delta$ T cell products.⁵⁸ While the peripheral blood accommodates approximately 1%–5% of the T cell population in the form of $\gamma\delta$ T cells, umbilical cord blood, another potential source of $\gamma\delta$ T cells, hosts less than 1%.^{21 59} As a result, the process of expansion of $\gamma\delta$ T cells has been studied preferentially for peripheral blood over umbilical cord blood, despite diminished chances of GvHD in the latter.^{29 50} Healthy donor apheresis can be used as an allogeneic source of V γ 9V δ 2 T cells, with isolated lymphocytes subsequently enriched preferentially, ex vivo, using aminobisphosphonates (eg, zoledronic acid or pamidronate) and cytokine supplementation (with agents such as IL-2), followed by depletion of contaminants (eg, $\alpha\beta$ T cells), using microbeads.⁶⁰ Alternatively, synthetic pAgs, like 2-methyl-3-butenyl-1-pyrophosphate or bromohydrin pyrophosphate, in combination with IL-2, have been used in different cancers for the activation of V δ 2 T cells, including solid cancers and hematological malignancies.^{29 61} To maintain an optimal level of cytotoxicity, these cells require coculture with K562 feeder cells for 10–14 days.⁶² Activation-induced cell death is a potential efficacy-limiting phenomenon seen in V γ 9V δ 2 T cells after prolonged exposure to pAgs, resulting in the production of exhausted cells.^{61 62} Recent approaches have addressed the numerical expansion constraints by adding a second phase of expansion of $\gamma\delta$ T cells using K562 feeder cell-derived artificial antigen-presenting cells, resulting in average pure-cell expansions as significant as >0.2 million-fold.^{21 61} Adoptive cell transfer has gained more relevance over in vivo activation of $\gamma\delta$ T cells because of its functional superiority, which is due to decreased anergy and exhaustion.²⁹ Allogeneic cell transfers are preferred in AML, despite the chances of alloreactivity, because of their cytotoxic dominance over their autologous counterparts, which are costly and time-consuming to harvest.⁶³ Adoptive cells also outrank autologous cells in terms of fitness, exhaustion, and overall activity.²⁹

$\gamma\delta$ T cell-based therapies in AML

Adoptive cellular therapy, bispecific engagers, and activating antibodies to harness $\gamma\delta$ T cells have been evaluated in AML. TCB-202-001, an allogeneic unmodified ex vivo-expanded V γ 9V δ 2 T cell therapy, was evaluated in a phase I study in patients with R/R AML. Three dose levels of 1×10^6 , 1×10^7 , and 1×10^8 cells/kg were evaluated using cells from matched or haploidentical donors. Patients underwent lymphodepletion with 25 mg/m² fludarabine for 5 days and 500 mg/m² cyclophosphamide for 2 days. Among two patients who received a low dose of TCB-202-001, 1 achieved a morphological leukemia-free state and one had stable disease. Among three patients at a higher dose level, one achieved complete remission (CR), one reported progressive disease, and one showed a significant blast count reduction on the 14th day. Patients with CR, morphological leukemia-free state, and stable disease received repeat infusions, without lymphodepletion; no toxicities, such as acute GvHD, ICANS, or CRS, were noted.^{64 65} A phase II trial with TCB-008 is currently ongoing in Europe.⁶⁶

Another trial evaluated the feasibility of delivering INB-100, an ex vivo-expanded donor-derived $\gamma\delta$ T cell therapy, to reduce the risk of relapse following allo-HSCT for R/R acute leukemias.^{67–69} Patients received reduced-intensity conditioning, followed by allo-HSCT from a haploidentical donor and a single infusion of donor-derived $\gamma\delta$ T cells within 5 days of neutrophil engraftment. Six patients received treatment at two dose levels of 1×10^6 cells/kg and 3×10^6 cells/kg. All patients had received one prior line of therapy except for one patient with ALL with prior exposure to chemotherapy and CAR-T cell therapy. Four patients developed grade 2 skin GvHD and three patients remained in remission after 1 year of follow-up.^{67 70} Several other clinical trials have evaluated adoptively transferred $\gamma\delta$ T cells in solid tumors and multiple myeloma.⁷¹ While the results from these trials were underwhelming, it is noteworthy that they were conducted in an era when there was only a rudimentary understanding of the biology of $\gamma\delta$ T cells.

More recently, novel activating antibodies and bispecific engagers have been evaluated to harness $\gamma\delta$ T cells in cancer. ICT01 is a first-in-class antibody that binds to BTN3A (CD 277) to activate the V γ 9V δ 2 T cell-mediated antitumor response.⁷² The ongoing EVICTION trial has shown robust pharmacodynamic effects, with rapid trafficking of activated V γ 9V δ 2 T cells into the tumor microenvironment and sustained partial responses in different solid tumors, including ipilimumab/nivolumab refractory melanoma (NCT04243499).⁷³ Azacytidine and venetoclax have been shown to potentiate the activity of ICT01, warranting evaluation in AML.⁷⁴ LAVA-051 is a humanized bispecific antibody that engages with CD1d and the V δ 2 TCR of the V γ 9V δ 2 T cells to mediate the killing of CD1d-positive tumor cells.⁷⁵ Early results from the ongoing dose escalation study demonstrated pharmacodynamic effects in terms of the activation of V γ 9V δ 2 T cells and invariant NK T cells, with no increase in

CRS-related cytokines, with predominantly grade 1/2 adverse events.^{76 77} The findings of relevant clinical trials have been summarized in [table 2](#).

Future directions

Although the existing information provides a promising insight into a potentially successful immunotherapy for AML, a vast realm remains to be explored. Further clarity may be attained when the allogeneic products in phase I/II clinical trials progress to phase III. Therapies may be developed that are modified on the basis of engineered $\gamma\delta$ T cells, enabling selective antitumor effects through gating mechanisms. Presently, clinical trials are primarily being conducted in R/R AML patients who may be shifted to diverse cohorts, including patients with early-stage AML. The reconstitution and persistence of $\gamma\delta$ T cells after transplantation must be a focus, as should enhancement of the persistence of terminally differentiated memory $\gamma\delta$ T cells, owing to their strong cytotoxic potential⁷⁸ (online supplemental figure S1). Studies on counteracting immune evasion by AML blasts and the mechanism of invigorating exhausted $\gamma\delta$ T cells could play a pivotal role in the therapeutic landscape of AML. In addition, the methods for enhancing the homing mechanism of $\gamma\delta$ T cells to the leukemia microenvironment and the exploration of the antileukemic properties of bone marrow-resident V δ 1 T cells could strengthen the immunotherapies and potentially broaden their application to patients with early-stage AML. $\gamma\delta$ T cells have unlocked new doors for AML immunotherapy; however, further research and investigation remain to be conducted before patients can benefit in the form of a large-scale off-the-shelf therapy.

CONCLUSIONS

$\gamma\delta$ T cells have emerged as a new modality of adoptive cellular therapy across cancers, particularly for R/R AML. Their unique biological characteristics, including a lack of alloreactivity, a lower probability of CRS and ICANS, and the capacity to influence the host immune microenvironment make them highly attractive candidates for off-the-shelf cellular therapy approaches. Reconstitution of these cells after alloHSCt may be suggestive of a favorable prognostic importance in AML patients. Susceptibility to functional exhaustion and relatively low persistence still remain the Achilles heel of $\gamma\delta$ T cells. Novel advances in expansion methods have led to various modified and unmodified $\gamma\delta$ T cell therapies to advance into clinical trials.

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